

Feature comparison

Two decks share one FSRS study engine underneath (vocabula's) but were built for different people: vocabula for one technically-engaged owner who wants every mechanic visible and tunable, riverbank for a general audience who should never have to think about scheduling internals.

FEATURE	VOCABULA	RIVERBANK
CONTENT		
Decks	Latin (3,034 words), Greek (983 words)	Latin (848), Greek (765), Italian (635)
Source material	Oxford freshers' reading-list vocabulary	OCR exam-board defined lists (Latin/Greek); a Duolingo-style study guide (Italian)
Categorisation	Latin: Parts I–VIII (syllabus order); Greek: none	Latin: 6 thematic tags (military/domestic/...); Greek: none; Italian: 39 topic sections
SCHEDULING		
Engine	FSRS-6, 21 fixed parameters	Same engine, unmodified
New words/day, target retention, weekly review day, reverse-direction gate	User-editable	User-editable
Longest interval cap	Editable	Removed — fixed default, no UI
Hesitation-adjusted grading	On/off toggle	Always on, no toggle
Easy-only clearing	On/off toggle	Always on, no toggle
Drill release (consecutive good days)	Editable, 0–14	Fixed at 3, no UI
Known-word head start / audit seeding	Full onboarding flow	Not present — every word starts unseen
STRUGGLING WORDS		
Leech/flag detection	5/8-repeat thresholds, manual + automatic flagging	Removed entirely, UI and scheduler both
Front-screen "Struggling words" list	Present	Removed
APPEARANCE		
Themes	7 (incl. Kanagawa, Everforest)	5 (dropped the two closest-looking)
Theme picker UI	Dropdown select	Tappable swatch row
Part filter UI	Chip row (fine at 8 parts)	Tappable bottom-sheet list with live counts (needed at 39 parts)
BACKUP & SYNC		
Mechanism	GitHub gist, id + personal access token	Short human sync code via a Cloudflare Worker
Setup required	Create a GitHub token and a gist by hand	Tap "Get a code," type it on the other device
Payload encryption	Plaintext (token is stripped, state itself is not encrypted)	AES-GCM, keyed from the sync code — server never sees readable data
Conflict handling	Named, counted, asks which copy to keep	Same logic, inherited unchanged

Manual export/import file	Present	Present
TRANSPARENCY & DOCS		
Settings help text	Full paragraphs explaining mechanism	One short sentence each, mechanism omitted
"What happens on its own" panel	Present — every background process listed	Removed
Manual	Technical — documents internals in depth	Deliberately shallow — usage only, no internals
Install guide	None	In-app + a standalone PDF
User feedback channel	None	In-app box, rate-limited, sent to a Worker inbox
TESTING		
Client-side logic tests	~69/65 checks (latin/greek), scheduler-focused	36 checks × 3 decks — data integrity, hardcoded-settings regression, FSRS invariants, sync logic
Backend tests	No backend to test	45 checks against the Worker (spam/injection guards, rate limits, sync round-trip)

What vocabula could borrow back

Riverbank isn't a strictly "improved" vocabula — most of its removals (leeches, the background panel, verbose settings text) are deliberate audience trade-offs vocabula's owner has explicitly said he doesn't want undone, since he prefers full visibility into the mechanism. But a few things below aren't audience trade-offs — they're just better engineering, worth adopting regardless of who the app is for.

- **Encrypt the gist payload.** Vocabula already strips the token before anything leaves the device, but the study-history content itself sits in the gist as plaintext — and vocabula's own handoff notes that "secret" gists are unlisted, not private. Riverbank's approach (derive an AES-GCM key from something the user holds, encrypt client-side, so the storage layer never sees readable data) would close that gap without changing anything about the gist workflow itself.
- **Group the settings screen.** Vocabula's ~15 fields are one long flat scroll. Riverbank's four labelled groups (Pace / Appearance / Backup & Sync / Danger zone) cost nothing structurally and would make a long settings page easier to scan, independent of how much detail stays in the help text.
- **Swap the theme dropdown for tappable swatches.** A visual pick beats a text dropdown for a visual choice — low-risk, self-contained change.
- **Keep the sheet-based part picker in reserve.** Latin's 8 parts are fine as chips today, but if the syllabus is ever re-split into more parts, riverbank's bottom-sheet list (with live word counts per row) is the version that won't need redoing later.
- **Assert data shape, not just scheduler correctness.** Vocabula's test suite is thorough on FSRS math but doesn't appear to check the word list itself (row shape, no empty terms, no duplicate ids). Riverbank's suite does, cheaply, and would have caught a bad row on the next data edit before it shipped.

Both apps share the same FSRS-6 core, the same single-file-per-deck philosophy, and the same conflict-naming logic on sync — none of that is duplicated effort, it's one engine serving two products.